Dmitrii **Altukhov**

Senior Python AI Engineer · PhD (Computer Science)

Amsterdam, The Netherlands

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Profile

ML Engineer with a PhD in computer science built around numerical optimization and a software engineering track record of building high-performance algorithms and scalable production AI systems. Background bridges algorithm-level performance engineering — CUDA/cuBLAS, C/C++, vectorized numerical methods, 10x algorithmic speedups — with distributed, event-driven production infrastructure. Strong foundations in linear algebra, optimization, and signal processing.

Performance & GPU	CUDA, cuBLAS, C/C++, numerical optimization, vectorization
Languages	Python, C/C++, Go, Rust (learning)
ML	PyTorch, deep learning, evaluation, LLM integration
Infrastructure	Kubernetes, Docker, Kafka, Redis, AWS, CI/CD, SQL
Quantitative	algorithms, linear algebra, statistics, signal processing

Experience

Creative Fabrica — *Amsterdam, the Netherlands*

ML Engineer → Senior ML Engineer — *Dec. 2023 – Jun. 2026*

- Owned the generative-AI serving stack end to end: **Kafka** ingress, **Redis** job queues, CPU pipeline workers, **Triton Inference Server** on GPU nodes, **Kubernetes** deployment, **Grafana** dashboards, metrics and alerting.
- Improved maintainability and observability by replacing in-house job-queue machinery with open-source **Redis**-backed queues.
- Redesigned a flaky service interface to **Triton**, moving it off base64 bytestring payloads onto the native protobuf transport.
- Developed an advanced color-quantization algorithm that became foundational for an industry-leading image vectorization pipeline.
- Shared on-call and contributed to the company's **Go** website backend.
- Productionized data scientists' model code; introduced type annotations and unit testing across the team's codebase.

Artificial Intelligence Research Institute (AIRI) — *Moscow, Russia*

Research fellow — *Feb. 2022 – Apr. 2023*

- Built and trained a deep learning model to predict speech from MEG data in **PyTorch**, meg_speech_decoding on GitHub.
- Developed a real-time **Python** pipeline for EEG timeseries prediction using Kalman filtering.

Higher School of Economics, Centre for Cognition and Decision Making — *Moscow, Russia*

Junior research fellow — *Feb. 2015 – Dec. 2021*

- Sped up by a factor of 10 the beamformer inverse solver in MNE-python by modifying the algorithm for vectorized computations.
- Profiled and accelerated various parts of my PhD codebase to enable Monte-Carlo statistics, using **CUDA** (cuBLAS) and **C**.
- Administered a laboratory **GPU** cluster running Linux.
- Led a 4-person team on real-time ML and GPU-accelerated 3D visualization in **Python**.

University of Montreal, CERNEC lab. — *Montreal, Canada*

Visiting researcher, temporary position — *Oct. – Dec. 2015, May 2016 – Dec. 2016*

- Co-developed an open-source **Python** package for GPU-accelerated brain data visualization, Visbrain.

Scientific Research Institute of System Analysis (SRISA RAS) — *Moscow, Russia*

Research assistant → Junior research fellow — *Jun. 2011 – Jan. 2015*

- Developed a droplet evaporation model in **C++**; tested **CUDA**-accelerated cluster software for jet-engine flow simulations.

Education

Ph.D. in Computer Science — *Moscow, Russia*

Higher School of Economics, Faculty of Computer Science — *Jan. 2016 – Nov. 2021*

GPA: 4.0; proposed an algorithmic framework for brain connectivity estimation; research published in the leading domain journals.

Specialist degree in Mechanics *(Masters equivalent)* — *Moscow, Russia*

Lomonosov Moscow State University, Department of Mechanics and Mathematics — *Sep. 2008 – Jun. 2013*

Thesis: "Enhancement and validation of LOGOS software for simulations of the reactive fluid flows."